Tora Matsuda

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Summary

Computer Science senior at UC Santa Cruz (B.S., Dec 2026) and undergraduate researcher building Kubernetes-based simulation and evaluation infrastructure for autonomous-vehicle safety. Ships full-stack products end to end (FastAPI, PostgreSQL, React, iOS) and serves as a course reader/tutor for algorithms and AI. Seeking full-time software engineering, ML infrastructure, or security roles starting 2027.

Education

University of California, Santa Cruz — Santa Cruz, CA
B.S. in Computer Science — *Sep 2022 – Dec 2026*

- **Coursework**: Analysis of Algorithms (CSE 102), Data Structures & Algorithms (101), Artificial Intelligence (140), Deep Learning (144), Database Systems – PostgreSQL (180), Network Programming (156), Parallel & Concurrent Programming (113), Distributed Systems (138), Computer Security (132), Cryptography (108)

Experience

AIEA Lab, UC Santa Cruz — Santa Cruz, CA
Undergraduate Researcher – AV Ethics & Safety — *Jan 2026 – Present*

- Built a CARLA 0.9.15 experiment pipeline on the Nautilus Kubernetes cluster (GPU Jobs, persistent volumes, Docker) measuring how AI decision latency changes autonomous-vehicle collision outcomes
- Ran a 144-trial controlled-delay grid (3 speeds × 6 time-to-collision × 8 delays) and replayed 18 measured LLM driving decisions (354 ms median latency) as CARLA trials
- Calibrated 1 ms braking physics to NHTSA and published-study targets (held-out error 0.8–11.6% vs. a 15% gate); authored a 22-parameter, source-cited table feeding the team's dilemma-window simulator
- Implemented seven-action maneuvers and paired inference/actuator-delay ablations with pedestrian actors (30 paired contrasts) for a team paper in preparation; earlier benchmarked PPO, DDPG, and SAC across CARLA and Gymnasium

UCSC Baskin Engineering — Santa Cruz, CA
Reader & Tutor – CSE 140 (Artificial Intelligence); Reader – CSE 102 (Analysis of Algorithms) — *Sep 2026 – Present*

- Grade programming projects, proof-based problem sets, and exams on search, MDPs/RL, recurrences, and dynamic programming; tutor students on AI algorithms and debugging Python agents

Reader & Tutor – CSE 101 (Data Structures & Algorithms) — *Jan 2026 – Apr 2026*

- Graded 500+ submissions across biweekly assessments for 100+ students; proctored exams and fielded questions on algorithm design and complexity

Projects

TasteMate – Live Music Social App | *React, FastAPI, PostgreSQL, Supabase, Capacitor* — Jun 2026 – Present

- Built React web and native iOS (Capacitor) clients on a FastAPI + PostgreSQL backend with 32 API routers, 33 schema migrations, and 1,100+ backend tests
- Engineered per-second FFT "feel" curves and a resonance score that match listeners by shared moments inside a song
- Shipped privacy-tiered live presence maps and listening rooms with Spotify PKCE, Apple, and email/phone OTP sign-in

LLM-Powered MITRE ATT&CK Command Generator | *Python, HuggingFace, RAG* — Sep 2025 – Jan 2026

- Built a multi-agent pipeline for adversarial command generation aligned with MITRE ATT&CK across 193+ techniques and 14 tactics
- Fine-tuned Llama 3.1 and Qwen via HuggingFace Transformers with RAG grounding to reduce hallucinated commands

100-Class Image Classification via Transfer Learning | *Python, PyTorch, DINOv2* — Apr 2026 – Jun 2026

- Fine-tuned DINOv2 ViT-B/14 in two stages with 16-pass test-time augmentation on ~1,000 training images
- Raised validation accuracy from ~63% to 82.79%; ranked 27th on Kaggle (score 0.836)

DAWLab – Collaborative Version Control for Music | *React, TypeScript, Electron, FastAPI* — Jan 2026 – Jun 2026

- Designed Git-style branching, merging, and rollback for DAW projects in an Electron app (Supabase, Clerk, Sentry)
- Adopted by a UCSC music production course; supports Ableton Live, Logic Pro X, and FL Studio

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Bash
Backend/Infra: PostgreSQL, FastAPI, Supabase, Docker, Kubernetes (Nautilus), CARLA, Git, CI/CD, TCP/UDP sockets, multithreading (C++ atomics, mutexes)
ML/AI: PyTorch, HuggingFace, LLM evaluation, RAG, RL (PPO, DDPG, SAC), DINOv2, scikit-learn, pandas, NumPy
Web/Mobile: React, Vite, Tailwind CSS, Electron, Capacitor (iOS)
Security: MITRE ATT&CK, Wireshark, nmap, hashcat, Kali Linux, cryptography, CTF competitions
Languages Spoken: English (Native), Japanese (Native/Bilingual), Spanish (Basic)